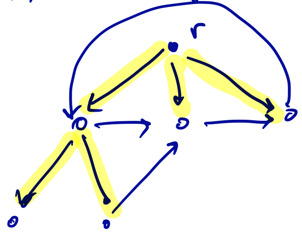
Min r-Arborescence

$G = (V, E)$, E directed edges.

$r \in V$, called the root.

A set of edges is called an r-arborescence if every node $v \in V \setminus \{r\}$ has exactly one arc (edge) terminating in v.



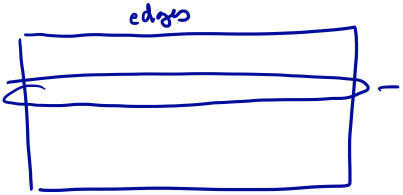
If each edge e has a length ℓ_e, the min length r-arb. problem is to find a min-length r-arb.

r-Arborescence and r-Cuts

Thm: The min length of an r-arborescence, when arc/edge e has length $\ell(e)$, is equal to the max # of r-cuts $C_1, \ldots, C_t$ s.t. no arc e is in more than $\ell(e)$ cuts

LP relaxation

Define a matrix C as follows:



— corresponds to a subset U of the nodes that defines a cut: $\emptyset \neq U \subseteq V \setminus \{r\}$.

$$\left.\begin{aligned} \min:\ & \ell^T x \\ \text{st}:\ & Cx \geq \mathbb{1} \\ & x \geq 0 \\ & \cancel{x_e \in \{0,1\}} \end{aligned}\right] \text{LP}$$

$$\boxed{\begin{aligned} \max:\ & \mathbb{1}^T y \\ \text{st}:\ & y^T C \leq \ell^T \\ & y \geq 0 \end{aligned}}$$

To prove thm, we show system is TDI. Must show max has an integral sol'n. $\forall \ell \in \mathbb{Z}^E$

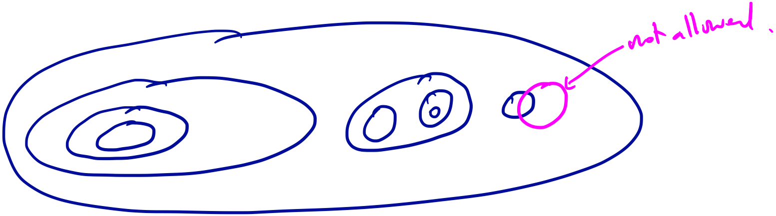
Thm: Both primal and dual have integral opt solutions, $\forall \ell \in \mathbb{Z}_+^E$ //

Proof — Laminarity

A collection $\mathcal{F}$ of subsets of S

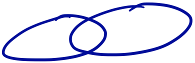
is called <u>Laminar</u> if $U_1, U_2 \in \mathcal{F}$

$$\Rightarrow U_1 \subseteq U_2 \text{ or } U_2 \subseteq U_1 \text{ or } U_1 \cap U_2 = \emptyset$$



(*) $\boxed{\begin{array}{ll} \max: & \mathbb{1}^T y \\ \text{st}: & y^T \underline{C} \le \ell^T \\ & y \ge 0 \end{array}}$

Let y be a maximizer of (*). If multiple maximizers, choose the one that makes $\underline{\sum_U y_U \cdot |U|^2}$ as large as possible.



<u>Claim</u>: $\mathcal{F} = \{U : y_U > 0\}$ is Laminar.

Let C' be the submatrix of C with rows corresponding only to those $U \in \mathcal{F}$.

Claim:

$$\begin{bmatrix} \max: \mathbb{1}^T y \\ \text{st}: y^T C \leq \ell^T \\ y \geq 0 \end{bmatrix} = \boxed{\begin{bmatrix} \max: \mathbb{1}^T y \\ \text{st}: y^T C' \leq \ell^T \\ y \geq 0 \end{bmatrix}}$$

pf: "$\geq$" is immediate.

"$\leq$" follows b/c we have a maximizer of LHS that has zeros for all rows not in C'.

$\boxed{\text{Claim: } C' \text{ is TU.}}$

This claim $\Rightarrow$ (**) has an integral solution

By prev. claim on equality of (*) & (**)

and b/c a sol'n to (**) can be made into a solution to (*) $\Rightarrow$ TDI

$\blacksquare$